COMPUTER ORGANIZATION AND ASSEMBLY LANGUAGE

LAB ASSIGNMENT NO. 02

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‘QUESTION 1’

Write an instruction to:

- Move value 5 into AX
- Move AX into BX
- Add BX and AX

‘ANSWER’

- **CODE EXPLAINED THROUGH COMMENTS:**

```
MOV AX, 5         ; Step 1: Put the value 5 into register AX
                  ; Now AX = 5
MOV BX, AX        ; Step 2: Copy the value from AX into BX
                  ; Now BX = 5, AX is still 5
ADD AX, BX        ; Step 3: Add BX to AX
                  ; AX = AX + BX
                  ; AX = 5 + 5 = 10
```

- **EXPLANATION:**

1. MOV AX, 5 → Loads the value **5** into register **AX**.
2. MOV BX, AX → Copies the value from **AX** into **BX**.
3. ADD AX, BX → Adds **BX** to **AX** and stores the result in **AX**.

After execution, **AX = 10** and **BX = 5**.

‘QUESTION 2’

Write Output:

- MOV AX, 5
- MOV BX, 3
- ADD AX, BX

- SUB AX, 2

what will be the final output of AX and BX?

‘ANSWER’

➢ **EXPLANATION OF EACH STEP:**

1. `MOV AX, 5`
→ AX becomes **5** (AX = 5)

2. `MOV BX, 3`
→ BX becomes **3** (BX = 3)

3. `ADD AX, BX`
→ AX = 5 + 3 = **8** (AX = AX + BX)

4. `SUB AX, 2`
→ AX = 8 − 2 = **6** (Subtracting 2 from AX, BX still 3)

➢ **FINAL OUTPUT:**

AX = 6 and **BX = 3**

‘QUESTION 3’

What are the differences between RAM and registers?

‘ANSWER’

➢ **RAM:**

- Keeps **programs and files open** while you are working.
- Bigger in size.
- Slower than registers.

➢ **REGISTERS:**

➢ Keeps **data that the CPU is using right now**.
➢ Very small.
➢ Very fast.

➢ COMPARISON TABLE:

FEATURE	RAM	REGISTERS
Definition	RAM is the main memory used to store programs and data temporarily.	Registers are very small memory units inside the CPU.
Location	Located outside the CPU (on the motherboard).	Located inside the CPU.
Speed	Slower than registers.	Fastest memory in the computer.
Size	Large capacity (GBs)	Very small capacity (few bytes).
Purpose	Stores data and programs currently in use.	Stores data that is being processed by the CPU.
Cost	Cheaper than registers	Very expensive.
Access Time	Takes more time to access.	Takes very little time (almost instant).
Example	8 GB RAM, 16 GB RAM	AX, BX, CX, DX

Thank You!!!!